

ROSHEN REJI

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SKILLS

AI / ML Core:	LLMs, RAG, Agentic AI, NLP, Computer Vision, Prompt Engineering, Model Evaluation
Frameworks:	LangChain, LangGraph, PyTorch, TensorFlow, Hugging Face, FastAPI, Flask
Vector & Retrieval:	FAISS, BM25, Hybrid Search, Embedding Models, Supabase
MLOps & Infra:	GCP, AWS (Amplify, Elastic Beanstalk), Firebase, Docker, Git
Languages:	Python, SQL, Java, C

EXPERIENCE

AI Developer

Mar 2025 – Jun 2025

Freelancer, Kerala, IN

- Engineered a deep learning crop disease classification system using ResNet-50 and TensorFlow, achieving **87%+ accuracy** across 10+ crop categories; deployed as a scalable REST API on GCP with automated JSON prediction outputs.
- Built a real-time AI inference API using Python and Flask on GCP, reducing manual diagnosis intervention with structured prediction workflows and request logging for monitoring and error tracing.
- Integrated Firebase for cloud-based storage and real-time retrieval of inference results, enabling persistent logging of model predictions across distributed field deployments.

PROJECTS

LucidCare — Multimodal Medical AI System

Mar 2026

YOLOv8 · DenseNet · LLM Reasoning · Flask · OCR · Supabase [IEEE Publication]

- Designed an end-to-end multimodal AI pipeline for chest X-ray interpretation using YOLOv8 for lesion localization, DenseNet for disease classification, and LLM-based reasoning to generate structured diagnostic outputs.
- Implemented Explainable AI (XAI) with visual-textual abnormality reasoning, improving model interpretability and enabling disease progression comparison across sequential reports.
- Built production-grade Flask APIs with OCR-driven report extraction and Supabase backend, supporting scalable async inference and cloud-based result storage.

Autonomous AI Analytics Platform

Dec 2025

LangGraph · FastAPI · Groq API · XGBoost · Docker · Python

- Architected a multi-agent agentic AI system using LangGraph with five specialized agents (Planner, Analytics, ML, Reflection, Report), enabling autonomous EDA, predictive modeling, and self-correcting workflow execution via reflection loops.
- Engineered stateful agent orchestration with conditional routing, retry logic, and quality evaluation scoring, improving pipeline reliability and enabling dynamic goal-driven dataset analysis at scale.
- Deployed Dockerized FastAPI backend with structured JSON outputs, real-time agent execution monitoring, and modular GitHub repo with documented architecture and run steps.

AstroChat — RAG-Powered Research Assistant

May 2025

RAG · LangChain · FAISS · BM25 · Meta-LLaMA · Flask · AWS

- Implemented a production RAG pipeline using hybrid retrieval (BM25 + FAISS vector search) with Meta-LLaMA APIs, improving semantic relevance and context-aware response quality for domain-specific queries.
- Designed chunking strategy, embedding model selection, and retrieval evaluation workflow, tracked retrieval quality and inference latency across iterations with documented experiment notes.
- Deployed full-stack AI system on AWS Amplify and Elastic Beanstalk with Flask backend, maintained modular codebase with PR-based GitHub workflow and reproducible deployment steps.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

Nov 2022 – May 2026

Muthoot Institute of Technology and Science, Kerala, India